

Bryan Chuya

Over in Packard Lab, we picked up the basics of hardware by hooking up simple circuits on breadboards. We played around with power supplies, LEDs, knobs (potentiometers), buttons, light sensors, and resistors. Along the way, we learned how to wire everything safely to avoid short circuits, and we even saw firsthand what happens when a resistor burns out from getting too much power. We also figured out that twisting the knob adjusts how sensitive the light sensor is.

After that, we plugged in an Arduino Nano ESP32 board and used Arduino Labs for MicroPython, with a little help from ChatGPT, to write code for a simple security setup. We quickly realized that being too vague with ChatGPT broke our code, so we started giving it our exact pin layout: D10 for the light sensor, D8 for the button, D3 for the green LED, and D5 for the yellow LED. In the end, our code turned the system on automatically when it got dark—turning on the green LED for safety and lighting up the yellow LED as an alarm whenever someone pushed the button.

For our final project, we hooked up a small motor (servo) and a button to the Arduino, writing MicroPython code so the motor would spin 180 degrees at the press of a button. Next, we tried out a rain sensor under different conditions, ranging from completely dry to touching a damp cloth. We started out measuring and writing down the sensor's voltage by hand using a multimeter, but soon wired it straight to the microcontroller to gather data on its own. Finally, we dumped the numbers into Google Sheets to plot a graph, which made it super easy to compare our manual notes against the automatic data and spot any slip-ups.